patterns of string vibration, they realized that their properties matched perfectly those of the hypothesized messenger particle of the gravitational force—the graviton. Although these “smallest bundles” of the gravitational force have, as yet, never been seen, theorists can confidently predict certain basic features that they must possess, and Scherk and Schwarz found these properties to be realized exactly by certain vibrational patterns. Based on this, Scherk and Schwarz suggested that string theory had failed in its initial attempt because physicists had unduly constrained its scope. String theory is not just a theory of the strong force, they proclaimed; it is a quantum theory that includes gravity as well.3

The physics community did not receive this suggestion with unbridled enthusiasm. In fact, Schwarz recounts that “our work was universally ignored.”4 The path of progress was already littered with numerous failed attempts to unite gravity and quantum mechanics. String theory had been shown wrong in its initial effort to describe the strong force, and it seemed to many that it was senseless to try to use the theory to pursue an even grander goal. Even more devastating, subsequent studies during the late 1970s and early 1980s showed that string theory and quantum mechanics suffered from their own subtle conflicts. It appeared that the gravitational force had, once again, resisted incorporation into the microscopic description of the universe.

Such was the case until 1984. In a landmark paper culminating more than a dozen years of intense research that had been largely ignored and often outright dismissed by most physicists, Green and Schwarz established that the subtle quantum conflict afflicting string theory could be resolved. Moreover, they showed that the resulting theory had sufficient breadth to encompass all of the four forces and all of matter as well. As word of this result spread throughout the worldwide physics community, particle physicists by the hundreds dropped their research projects to launch a full-scale assault on what appeared to be the last theoretical battleground in the ancient quest to understand the deepest workings of the universe.

I began graduate school at Oxford University in October 1984. Although I was excited to be learning about the likes of quantum field theory, gauge theory, and general relativity, there was a pervasive feeling among the older graduate students that there was little or no